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SEQUENTIAL SAMPLER SEQ47/50-RV

Features

- Stainless steel housing for **outdoor measurements**; **19" rack version** for installation in **measuring stations**
- Self-explanatory menu-guided operation by means of 3 keys:
Start and stop by hand
Start and stop automatic
Interval mode
- Data storage on **USB stick (data logger function)**
- **Bavarian-Hessian protocol / remote control**
- Controlling of **operating-m³/h** by orifice
- Protection of stored data against power failure, real time clock
- Impactor inlets with exchangeable jets (8 pieces)
for **PM10 – PM4,0 – PM2,5 – PM1,0**
- **TSPM Inlet** according to **VDI 2463 parts 5 and 8**
- Impactor inlet with **ozone denuder for PAHs (BaP)** according to **CEN EN 15544**
- **Temperature measurement directly downstream the filter**
- **Cooled inlet system by sheath air and covered sampled filters** inside the magazine, no losses of particulate volatile material
- Sampling tube (stainless steel) with large inside diameter, **no wall losses**



**PM10 and PM2,5 STANDARD
REFERENCE SAMPLER
according to CEN EN 12341:2014**

- Use of filters with diameters of **47 mm** and **50 mm**

Description

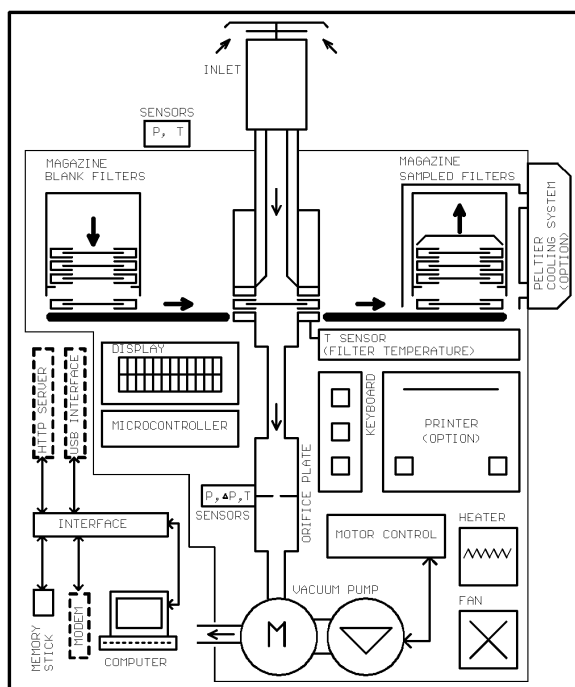
The sequential sampler **SEQ47/50** is available in 2 versions: as outdoor sampler (see picture on the front page) and as station sampler for the installation in a 19" rack with extended sampling tube (**maximum: 3 m**).

The housing of the outdoor sampler consists of stainless steel sheet metal of 1,5 mm thickness with a lockable door. The further solid construction guarantees a maintenance-free operation of the sampler for a long operating period.

The outdoor sampler can be operated at ambient temperature of approx. $-30\text{ }^{\circ}\text{C}$ up to more than $50\text{ }^{\circ}\text{C}$ at all weather conditions.

The **magazines** for the blank and sampled filters are able to load 17 filter holders, each. By covering the upper filter holder and superimposition of the following filter holders inside the magazine for the sampled filters the air volume above each filter is so small that particulate volatile material cannot evaporate into the air. Additionally, the complete sampling system up to the filter is cooled by sheath air. By these measures a reliable determination of the particle masses collected on the filters is guaranteed also after a complete sampling period.

The **filter holders (POM)** are capable to take in filters with diameters of 47 mm and 50 mm as well. Filter holders for filters with a diameter of only 47 mm are also available.



BLOCK DIAGRAM SEQ47/50

The **flow rates** are controlled in terms of **operating- m^3/h** in compliance with basic physical principles by means of a temperature- and pressure-compensated orifice according to Bernoulli's law and by conversion to ambient conditions according to Boyle-Mariotte's law.

The flowrates and sampled volumes are displayed and recorded (**USB-stick**) in terms of **operating- m^3/h** resp. **operating- m^3** (related to ambient conditions, each) and

additionally recorded in terms of **standard- m^3/h** resp. **standard- m^3** (related to $0\text{ }^{\circ}\text{C}$ and 760 mmHg, each).

The flow rates as well as the temperature and pressure sensors can be easily re-calibrated by means of the 3 front keys.

The inside diameter of the sampling tube (**stainless steel**) is 27 mm.

Inlets

- Use of all **PMX Inlets** for the flow rates of **1,0 - 2,3 - m^3/h**
- **PM10 and PM2,5 measurements according to CEN EN 12341**
- **Total dust measurement according to VDI 2463 Parts 5 and 8**
- **PAH (BaP) measurement according to CEN EN 15549 and Directive 2004/107/EC**
- **Measurements of heavy metals according to CEN EN 14902**
- **Measurements of soot (EC/OC) according to VDI 2465 and CEN/TR 16243)**

Technical Data

Flow rates

controlled 1,0 – 2,3 m^3/h
Deviation from the set point: $< 2\%$

Sampling time

minimum 1 h – maximum 168 h per filter

Breaks (interval mode)

minimum 1 h – maximum 168 h

Power supply

230 V, 50/60 Hz

Consumption

approx. 300 VA

Filter diameter

47 – 50 mm

Diameter of active filter area

approx. 38 – 40 mm

Dimensions

Width 482 mm
Depth 310 mm
Height with inlet 1,58 m

Weight

approx. 60 kg (transportable by casters)

Noise level according to DIN 2058

in a distance of 8 m: $< 35\text{ dBA}$

Subject to alterations Ed. 03/16